# Gold Future/ETF Analysis

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Table 1: New Data: A summary of the regression coefficients and measures of goodness of fit for regressing one-day returns of 1, 2, 6 and 12-month futures on 1-day returns of spot gold.

| **Maturity** | **Slope** | **Intercept** | $R^2$ | **RMSE** |
|---|---|---|---|---|
| 1-Month | 0.9973 | — | 0.9461 | — |
| 2-Month | 1.0001 | — | 0.9439 | — |
| 6-Month | 1.0042 | — | 0.9369 | — |
| 12-Month | 0.9462 | — | 0.8286 | — |

*Slopes are very close to 1, apart from the 12-month contract. This may be a result of the lower liquidity of the longer contract and the smaller impact of daily gold price movements on longer-dated futures.*

Table 2: Leung & Ward (2015) Data: Regression coefficients for one-day futures returns on one-day spot gold returns.

| **Maturity** | **Slope** | **Intercept** | $R^2$ | **RMSE** |
|---|---|---|---|---|
| 1-Month | 0.94314 | $2.4191 \times 10^{-5}$ | 0.80947 | 0.00530 |
| 2-Month | 0.94301 | $1.0000 \times 10^{-5}$ | 0.80911 | 0.00530 |
| 6-Month | 0.94348 | $3.1797 \times 10^{-5}$ | 0.80934 | 0.00530 |
| 12-Month | 0.94358 | $-5.2333 \times 10^{-5}$ | 0.80984 | 0.00529 |

*The slopes are also very close to 1, but consistently lower than the newer data. The $R^2$ values are also significantly lower than the newer data. This may be due to the smaller sample size of the older data and lower liquidity.*

*a*

Table 3: New Data: Slopes and $R^2$ from the regressions of futures returns versus gold returns over different holding periods.

| | **Days** | **1-Month** | **2-Month** | **6-Month** | **12-Month** |
|---|---|---|---|---|---|
| **Slope** | 1 | 0.99730 | 1.00010 | 1.00420 | 0.94620 |
| | 5 | 1.01040 | 1.01320 | 1.01250 | 0.98780 |
| | 10 | 1.00320 | 1.00600 | 1.00390 | 0.98840 |
| | 15 | 1.00370 | 1.00500 | 1.00340 | 0.99220 |
| $R^2$ | 1 | 0.94610 | 0.94390 | 0.93690 | 0.82860 |
| | 5 | 0.98080 | 0.98100 | 0.97820 | 0.95350 |
| | 10 | 0.99020 | 0.98960 | 0.98630 | 0.97370 |
| | 15 | 0.99310 | 0.99240 | 0.98940 | 0.97930 |

*The slopes and $R^2$ values generally increase with longer holding periods but decrease with longer-dated contracts.*

Table 4: Leung & Ward (2015) Data: Slopes and $R^2$ from regressions of futures returns vs. gold returns over different holding periods.

| | **Days** | **1-Month** | **2-Month** | **6-Month** | **12-Month** |
|---|---|---|---|---|---|
| **Slope** | 1 | 0.94314 | 0.94301 | 0.94348 | 0.94358 |
| | 5 | 0.99805 | 0.99752 | 0.99757 | 0.99715 |
| | 10 | 1.01075 | 1.01020 | 1.00970 | 1.00841 |
| | 15 | 0.99448 | 0.99461 | 0.99431 | 0.99345 |
| $R^2$ | 1 | 0.80947 | 0.80911 | 0.80934 | 0.80984 |
| | 5 | 0.97154 | 0.97158 | 0.97194 | 0.97165 |
| | 10 | 0.98516 | 0.98530 | 0.98572 | 0.98550 |
| | 15 | 0.98704 | 0.98691 | 0.98725 | 0.98713 |

The out of sample period is defined as starting on January 1, 2024 and is approximately 10% of the data. The RMSEs are calculated in dollar terms assuming a notional investment of $1000 in the spot gold price. The weights are calculated using a constrained least squares regression that forces the weights to sum to one. The GLD RMSE is $6.00 for the out of sample period.

Table 5: New Data: Weights and in/out of sample RMSEs for portfolios of 1, 2, 3, and 4 futures contracts.

| **Portfolio** | $w_0$ | $w_1$ | $w_2$ | $w_6$ | $w_{12}$ | $\mathbf{RMSE_{in}}$ | $\mathbf{RMSE_{out}}$ |
|---|---|---|---|---|---|---|---|
| 1-m | 0.00256 | 0.99743 | | | | $2.65 | $5.15 |
| 2-m | 0.00676 | | 0.99324 | | | $3.67 | $5.75 |
| 12-m | 0.04509 | | | | 0.95491 | $18.84 | $17.95 |
| 1-m, 2-m | 0.00168 | 1.21224 | -0.21393 | | | $2.61 | $5.06 |
| 1-m, 6-m | 0.00165 | 1.04753 | | -0.04918 | | $2.61 | $4.88 |
| 1-m, 12-m | 0.00189 | 1.01446 | | | -0.01635 | $2.66 | $4.95 |
| 2-m, 6-m | 0.00330 | | 1.23075 | -0.23405 | | $3.18 | $4.63 |
| 2-m, 12-m | 0.00310 | | 1.09461 | | -0.09770 | $3.26 | $4.99 |
| 6-m, 12-m | 0.00446 | | | 1.76090 | -0.76537 | $6.07 | $12.15 |
| 1-m, 2-m, 6-m | 0.00156 | 1.14088 | -0.11333 | -0.02911 | | $2.60 | $4.94 |
| 1-m, 2-m, 12-m | 0.00170 | 1.23840 | -0.24373 | | 0.00363 | $2.64 | $4.99 |
| 1-m, 6-m, 12-m | 0.00197 | 1.10522 | | -0.17107 | 0.06388 | $2.62 | $4.54 |
| 2-m, 6-m, 12-m | 0.00327 | | 1.22084 | -0.21809 | -0.00602 | $3.21 | $4.46 |
| 1-m, 2-m, 6-m, 12-m | 0.00186 | 1.21485 | -0.13541 | -0.14322 | 0.06191 | $2.62 | $4.62 |

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Table 6: Leung & Ward (2015) Data: Weights and in/out of sample RMSEs for portfolios of 1 and 2 futures contracts.

| **Portfolio** | $w_0$ | $w_{short}$ | $w_{long}$ | $\mathbf{RMSE_{in}}$ | $\mathbf{RMSE_{out}}$ |
|---|---|---|---|---|---|
| 1-m | -0.01071 | 1.01071 | | $6.63 | $3.34 |
| 2-m | -0.04835 | 1.04835 | | $12.22 | $6.12 |
| 12-m | -0.06842 | 1.06842 | | $15.11 | $5.71 |
| 1-m, 2-m | -0.00088 | 1.27315 | -0.27227 | $6.27 | $2.79 |
| 1-m, 6-m | -0.00171 | 1.23899 | -0.23729 | $6.27 | $2.98 |
| 1-m, 12-m | -0.00021 | 1.19336 | -0.19315 | $6.29 | $3.00 |
| 2-m, 6-m | -0.04079 | 5.06602 | -4.02523 | $10.27 | $9.37 |

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Table 7: New Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

| **(L)ETF** | **Leverage ($\beta$)** | **Slope** | **Intercept** | **t-stat** | **p-value** | $R^2$ | **RMSE** |
|---|---|---|---|---|---|---|---|
| GLD | 1 | 1.00061 | $-8.8911 \times 10^{-6}$ | 0.25954 | 0.79523 | 0.97140 | 0.00185 |
| UGL | 2 | 2.02088 | $-1.9483 \times 10^{-4}$ | 2.77734 | 0.00552 | 0.96438 | 0.00349 |
| GLL | -2 | -2.01429 | $1.8890 \times 10^{-4}$ | -1.92313 | 0.05457 | 0.96497 | 0.00345 |
| DGLD | -2 | -1.97565 | $-8.4896 \times 10^{-5}$ | 0.46841 | 0.63954 | 0.40656 | 0.02098 |
| UGLDF | 3 | 3.00632 | $4.5874 \times 10^{-4}$ | 0.05822 | 0.95358 | 0.25518 | 0.04463 |

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Table 8: Leung & Ward (2015) Data: Regression coefficients for one-day returns of (L)ETFs versus spot gold.

| **(L)ETF** | **Leverage ($\beta$)** | **Slope** | **Intercept** | **t-stat** | **p-value** | $R^2$ | **RMSE** |
|---|---|---|---|---|---|---|---|
| GLD | 1 | 1.00540 | $-1.6427 \times 10^{-6}$ | 1.42692 | 0.15382 | 0.98060 | 0.00163 |
| UGL | 2 | 2.00572 | $-1.3107 \times 10^{-4}$ | 0.73319 | 0.46357 | 0.97930 | 0.00337 |
| GLL | -2 | -2.00556 | $-1.0550 \times 10^{-4}$ | 0.67089 | 0.50240 | 0.97673 | 0.00358 |
| UGLD | 3 | 2.99358 | $-1.9860 \times 10^{-4}$ | 0.45211 | 0.65133 | 0.98484 | 0.00421 |
| DGLD | -3 | -2.97528 | $-1.6912 \times 10^{-5}$ | 0.97442 | 0.33019 | 0.95256 | 0.00752 |

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The t-statistics far from zero for UGL and GLL indicate that the slopes are statistically significantly different from the specified leverage ratios.

Table 9: New Data: Weights and in/out of sample RMSEs for portfolios replicating leveraged benchmarks.

| **Benchmark** | **Portfolio** | $w_0$ | $w_1$ | $w_2$ | $w_6$ | $w_{12}$ | $\mathbf{RMSE_{in}}$ | $\mathbf{RMSE_{out}}$ |
|---|---|---|---|---|---|---|---|---|
| 2x Leveraged | 1-m | -1.63329 | 2.63329 | | | | \$59.08 | \$330.19 |
| | 2-m | -1.62266 | | 2.62266 | | | \$56.34 | \$334.55 |
| | 6-m | -1.57547 | | | 2.57547 | | \$46.04 | \$356.16 |
| | 12-m | -1.50287 | | | | 2.50287 | \$46.58 | \$375.04 |
| | 1-m, 2-m | -1.57511 | -10.84619 | 13.42130 | | | \$50.91 | \$352.51 |
| | 1-m, 6-m | -1.55949 | -1.36860 | | 3.92809 | | \$45.60 | \$363.58 |
| 3x Leveraged | 1-m | -4.03993 | 5.03993 | | | | \$188.14 | \$1035.66 |
| | 2-m | -4.02060 | | 5.02060 | | | \$183.07 | \$1042.01 |
| | 6-m | -3.92928 | | | 4.92928 | | \$161.06 | \$1076.52 |
| | 12-m | -3.78922 | | | | 4.78922 | \$147.86 | \$1116.50 |
| | 1-m, 2-m | -3.86549 | -35.37385 | 40.23934 | | | \$165.29 | \$1085.27 |
| | 1-m, 6-m | -3.81478 | -7.16800 | | 11.98279 | | \$149.22 | \$1110.26 |

Table 10: Leung & Ward (2015) Data: RMSEs for portfolios replicating leveraged benchmarks.

| **Benchmark** | **Portfolio** | $\mathbf{RMSE_{in}}$ | $\mathbf{RMSE_{out}}$ |
|---|---|---|---|
| +2x (UGL) | 1-m | \$153.20 | \$41.49 |
| | 2-m | \$140.41 | \$49.57 |
| | 6-m | \$138.63 | \$47.46 |
| | 12-m | \$134.51 | \$48.30 |
| +3x (UGLD) | 1-m | \$555.49 | \$111.51 |
| | 2-m | \$529.91 | \$125.99 |
| | 6-m | \$526.58 | \$122.33 |
| | 12-m | \$518.97 | \$123.83 |